

The 45-Minute CRM: A Hybrid AI Development Debrief

Building a Full-Stack  System using Claude-Flow Swarms & Direct Assistance

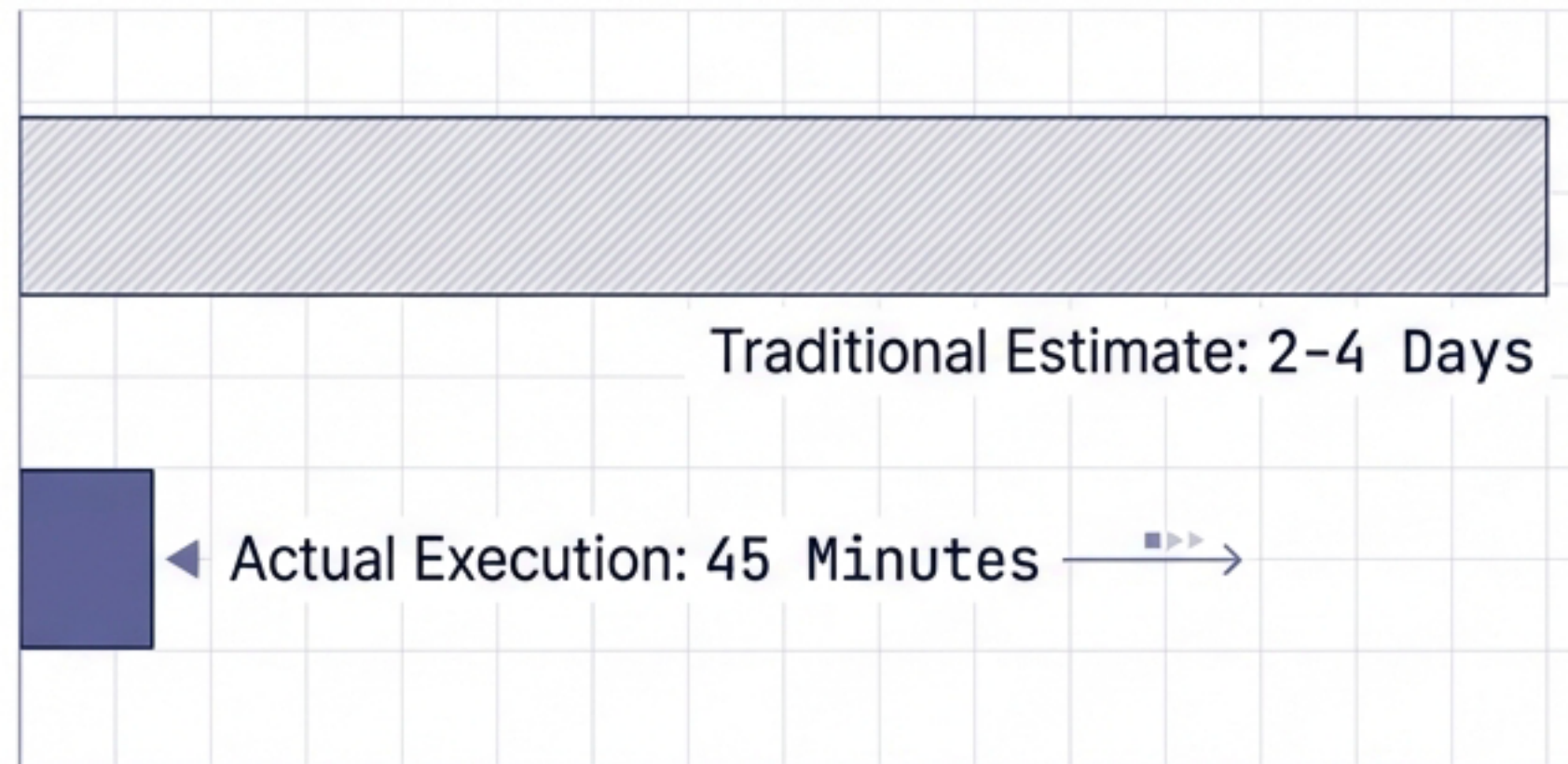
```
> initiating_swarm_sequence...
> agents_active: 5
> generating: src/types/index.ts [DONE]
> generating: src/storage/Store.ts [DONE]
> testing: 40/40 passed (100%)
> build_complete: 45m 12s
```

Customers Deals Analytics	Deals Pipeline		
	New Lead	Contacted	Proposal
	TechCorp SaaS Upgrade - \$50k Acme Inc Consult - \$12k		

From Zero to -Ready in Under an Hour

Challenging the standard velocity of software delivery.

THE COMPARISON



THE DELIVERABLES

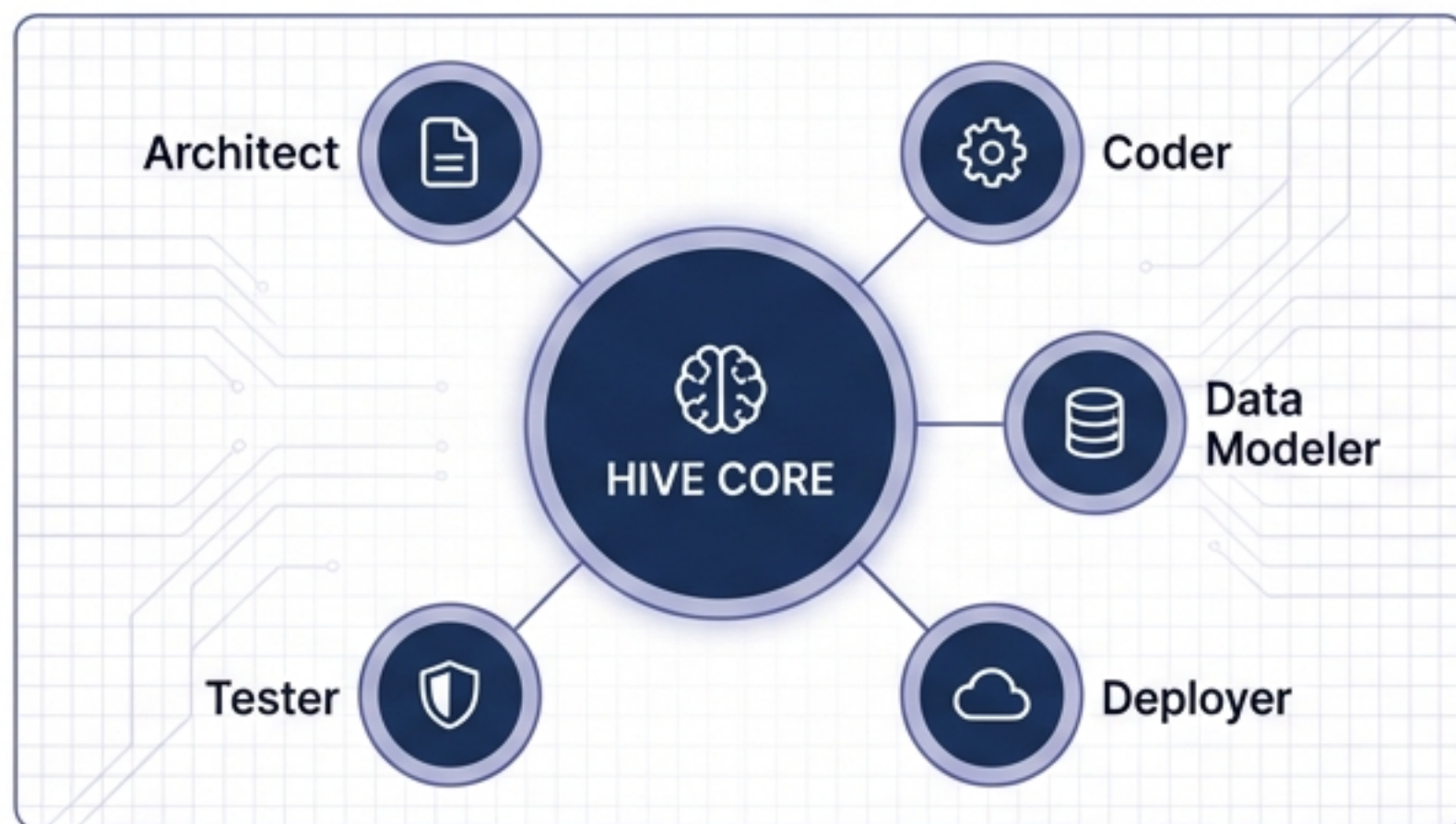
- ✓ **Backend Architecture:** REST API with 18 endpoints across 3 resources (Customer, Deal, Interaction).
- ✓ **Automated Documentation:** Interactive Swagger UI (OpenAPI 3.0) generated automatically.
- ✓ **Frontend Experience:** Modern, responsive SPA with Drag-and-Drop Kanban board.
- ✓ **Quality Assurance:** 40 Unit Tests with 100% Pass Rate.

 **Core Insight:** The result was achieved not by coding faster, but by parallelizing the workload through 'Swarm Orchestration' and refining via 'Direct Assistance' 

The Methodology: Two Distinct Modes of Execution

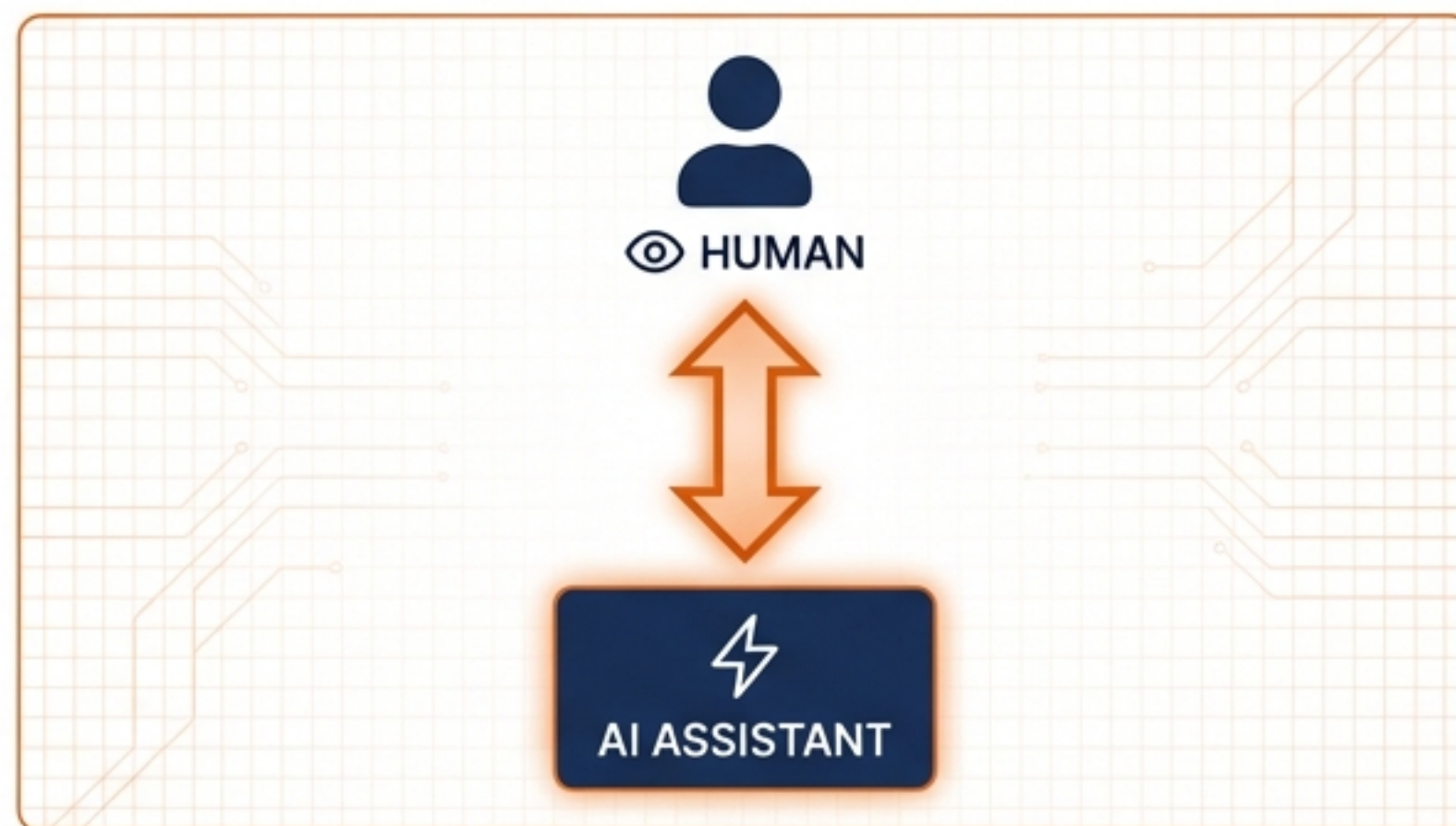
Success required alternating between 'Hive-Mind' breadth and 'Sniper' depth.

Mode 1: Swarm Orchestration (The Hive)



- Multiple instances running in parallel
- Background execution (`run_in_background: true`)
- Shared 'Hive-Mind' memory state
- Best for: Large, parallelizable workloads

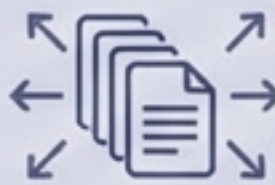
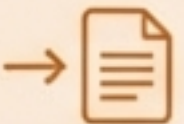
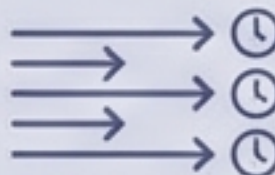
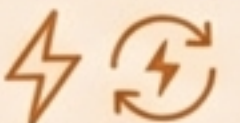
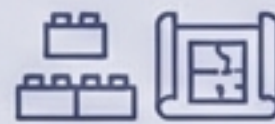
Mode 2: Direct Assistance (The Sniper)



- Single instance, single thread
- Synchronous / Immediate edits
- High-speed iterative feedback loop
- Best for: Debugging, Runtime fixes, Config

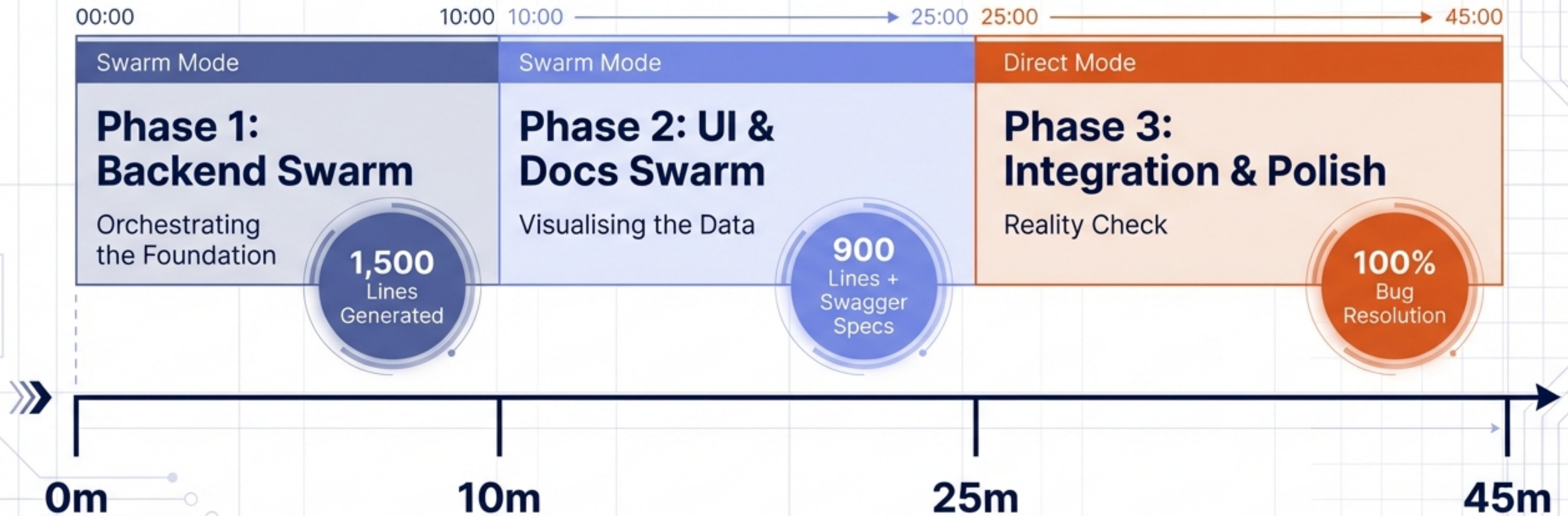
Optimising Agent Deployment via Strategic Selection

Success depends on aligning the execution mode with the task characteristics.

Criteria	Swarm Mode (The Hive)	Direct Mode (The Sniper)
Task Size	Large (10+ files) 	Small (1-4 files) 
Concurrency	High Parallelism (Async) 	Linear / Sequential (Sync) 
Iteration Speed	High throughput, slower startup 	Immediate feedback loop 
Ideal Use Case	New Feature Development & Architecture 	Bug Fixes, Config & Polish 

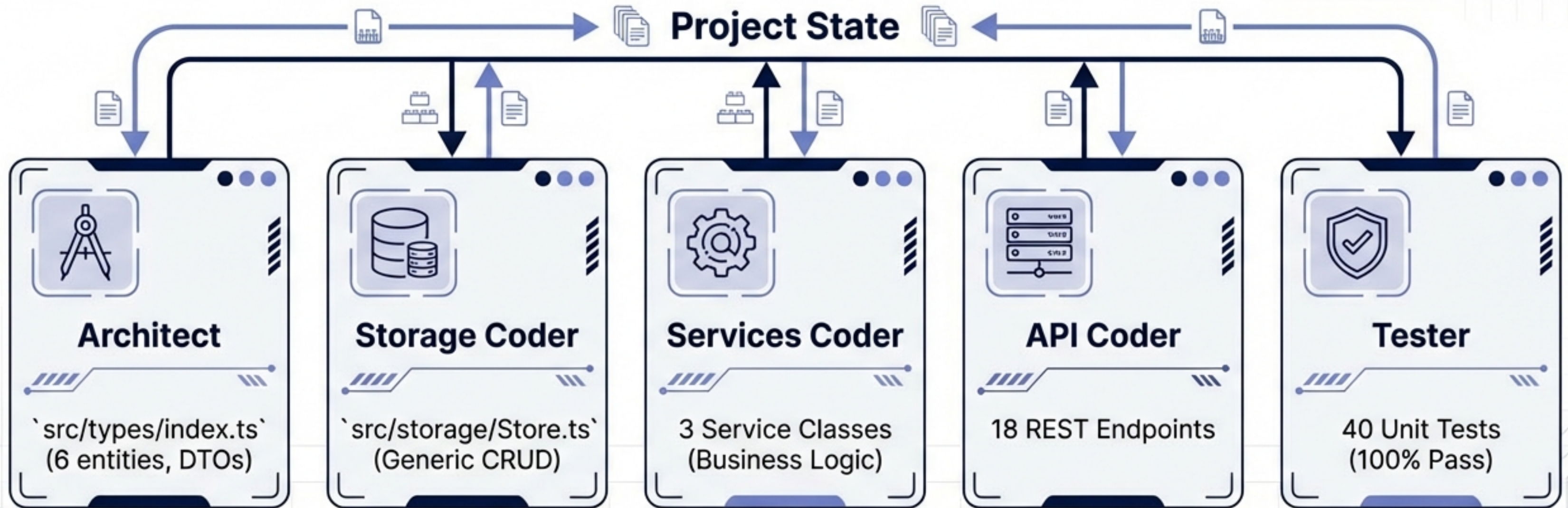
 *Agent overhead is worth it for complexity; speed is king for iteration.* 

The 45-Minute Sprint Timeline



Phase 1: Orchestrating the Foundation

Time: 10 Minutes | Mode: Swarm Orchestration



Efficiency Gain: Sequential estimate was 30+ minutes.
Swarm parallel execution reduced this to ~10 minutes.

Phase 2: Visualising the Data

Time: 15 Minutes | Mode: Swarm Orchestration

Agent Roster



Visual Outcome



Auto-Generated Documentation
(`/api-docs`)

Phase 3: The Integration Reality Check

Time: 20 Minutes | **Mode:** Direct Assistance (The Sniper)

Problem / Resolution		
ISSUE	ROOT CAUSE	THE FIX
<code>`TypeError: this.deals.reduce is not a function`</code>	API returned <code>{success, data}`</code> object, Frontend expected raw Array.	Updated <code>APIClient.ts</code> to unwrap the response (<code>res.data</code>).
Field Name Mismatches (undefined values)	Contract breach: API used <code>`camelCase`</code> , Frontend used <code>`snake_case`</code> .	Global regex find/replace to standardize naming.
HTML Rendering Error: Invalid Token	UUID strings passed to <code>`onclick`</code> handlers without quotes.	Added escaped quotes: <code>onclick='edit("\${id}")'</code> .

****Why Direct Mode?****
High-speed feedback loops were required to debug runtime errors that static analysis missed. Swarms are too slow for this granular iteration.

Architecture Delivered: System Topology

QA & Documentation

Jest Unit Tests

Swagger UI

Frontend (Client)

Vanilla JS

CSS Grid Layout

Drag-and-Drop API

public/js/app.js

REST API / JSON

Backend (Server)

Express App

Service Layer
(Business Logic)

Data Store
(File System)

src/services/

src/storage/

Stats

Total Files:
19

Total Lines of Code:
~3,900
JetBrains Mono

Tech Stack:
**TypeScript /
Node.js**

A Feature-Complete CRM System



Deal Pipeline

6-stage Kanban board with drag-and-drop capability and real-time value tracking.



Customer Management

Full CRUD operations, search functionality, and status filtering.



Interaction Logging

Tracking for Calls, Emails, Meetings, and Notes.



Analytics Dashboard

KPIs including Customer Count, Deal Count, and Total Pipeline Value.



Company Profiles

Detailed tracking of industry, company size, and physical address data.

Quantifying the Efficiency Gain

Velocity

45 Mins

Total Time

vs Traditional: 2-4 Days

Volume

3,900

**Lines of Code
Generated**

Backend (TS): ~1,500
Frontend: ~1,900
Tests: ~500

Quality

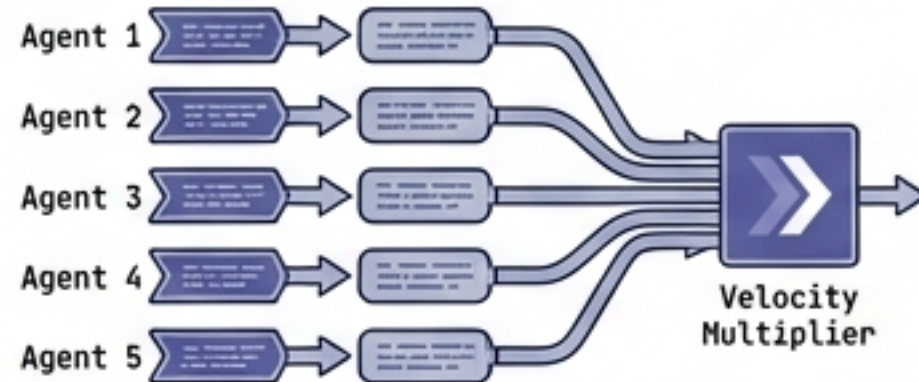
100%

Test Pass Rate

Storage Tests: 22
Service Tests: 18
Total Coverage: 40 Tests

Success Drivers: Why the Hive-Mind Worked

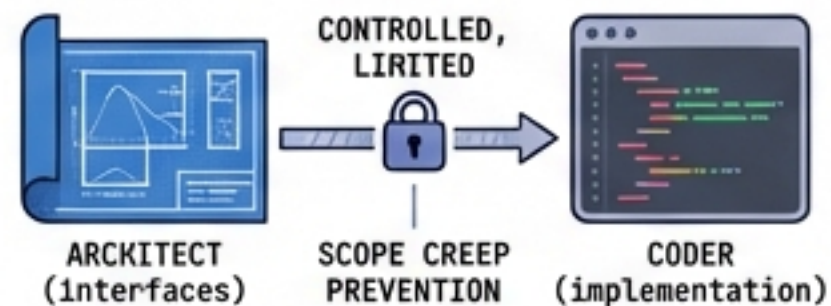
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Parallel Agent Execution

Spawning all agents in a single message was the primary velocity multiplier. Five streams of work happened simultaneously.

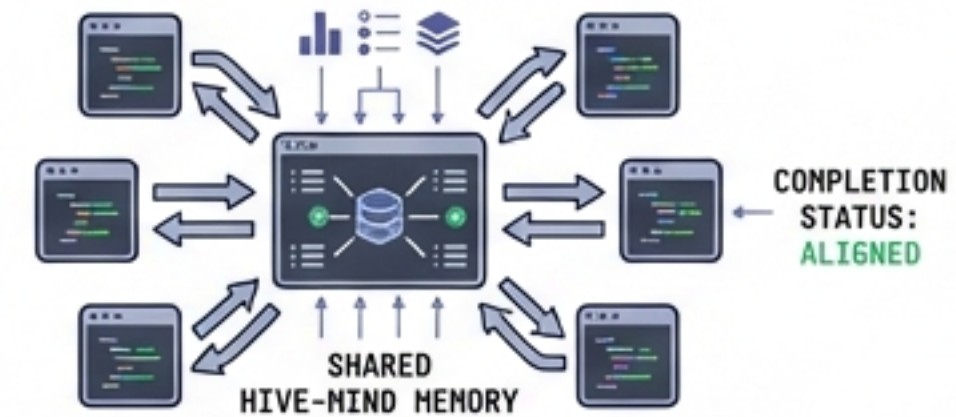
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Strict Role Specialization

Separating the 'Architect' (interfaces) from the 'Coder' (implementation) prevented scope creep and ensured structural integrity.

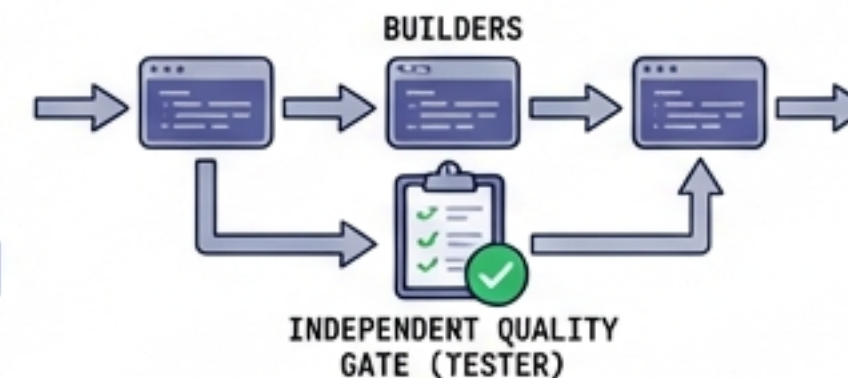
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Hive-Mind Memory

Agents utilized a shared state to align on architecture decisions and track completion status across the swarm.

4

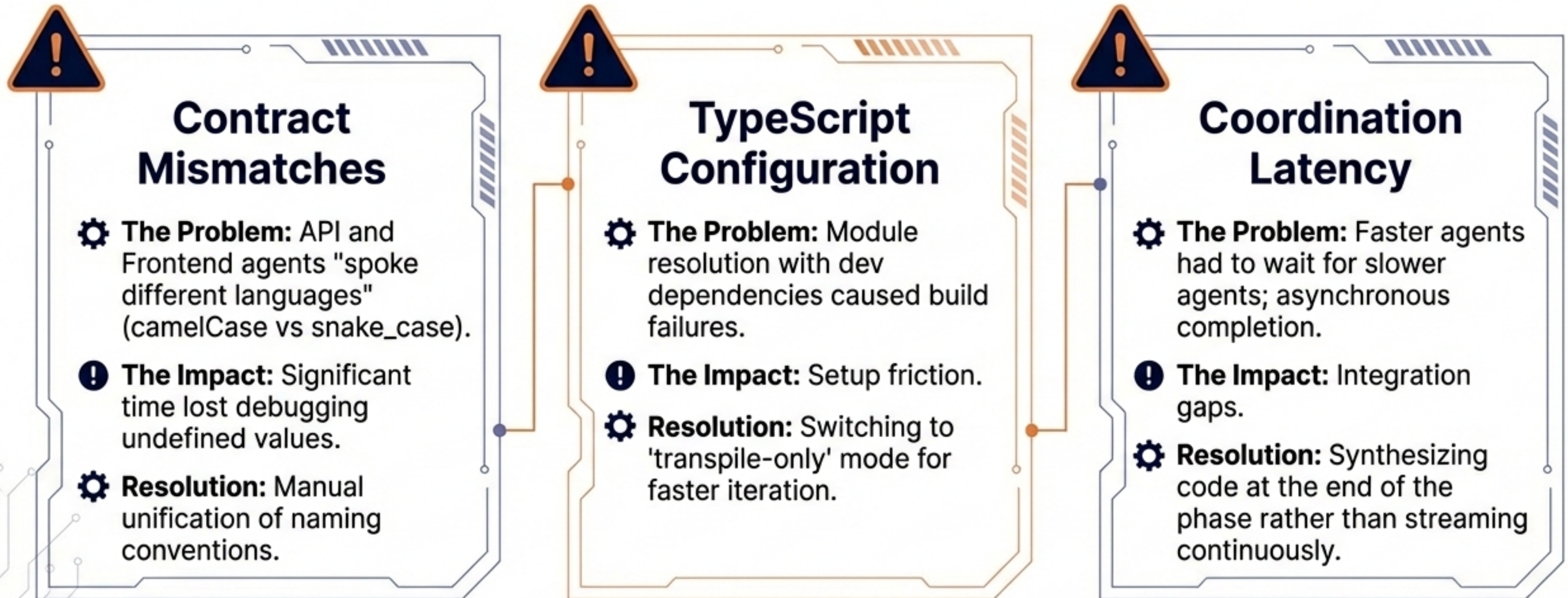


Independent Quality Gate

A dedicated 'Tester' agent ran in parallel, ensuring code validity without slowing down the builders.

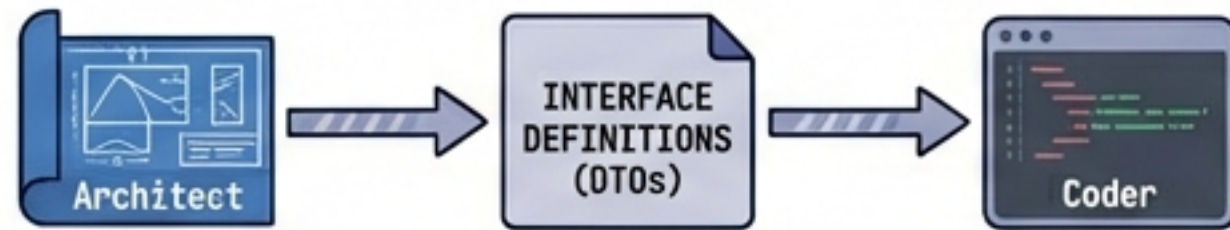
Friction Points & Challenges

Where AI development still requires human oversight.



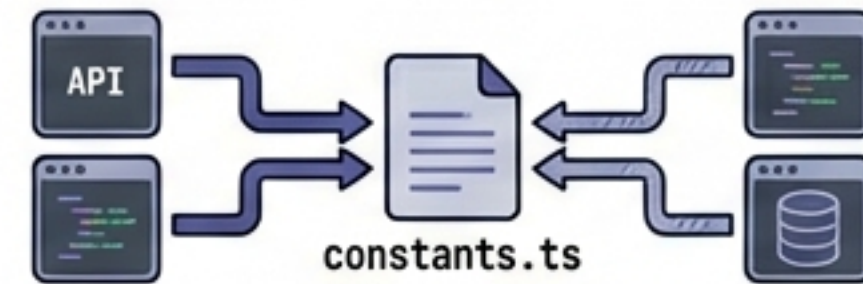
Optimising the Next Sprint

Recommendations for future high-velocity projects.



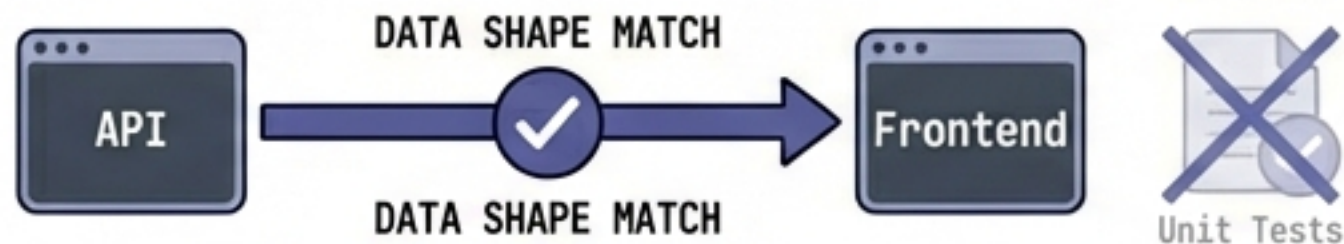
Define Contracts First

The Architect agent must output strict interface definitions (DTOs) *before* any coding agents begin work to ensure.



Single Source of Truth

Create a shared `constants.ts` file for field names and status values immediately to prevent casing bugs.



Integration Over Unit

Move beyond unit tests. Include integration tests that verify the API and Frontend data shapes match.

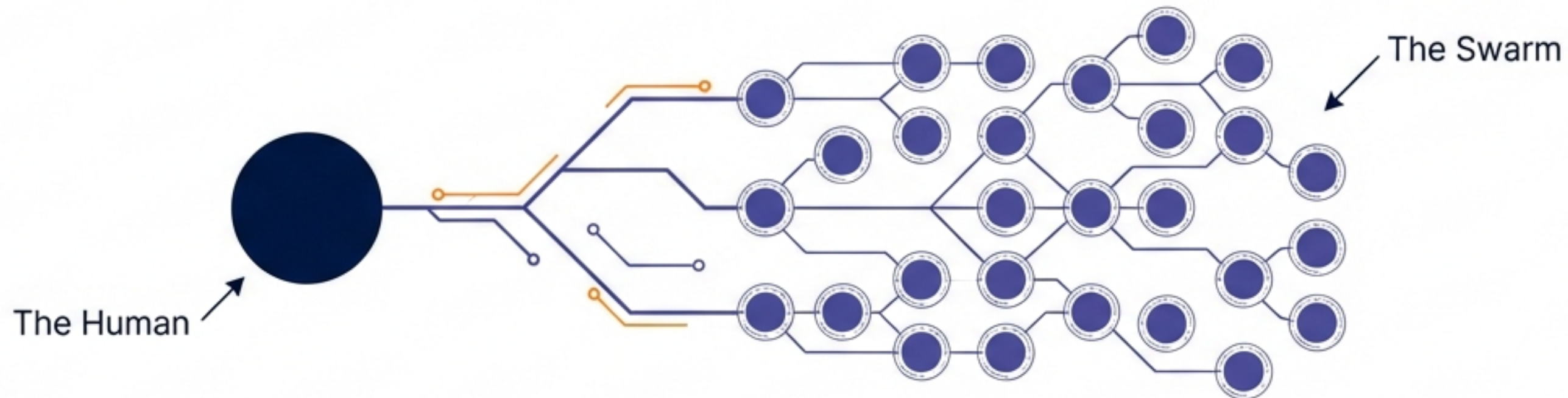


Incremental Verification

Implement a 'stop-and-check' gate between Backend and Frontend phases to catch errors before they cascade.

The Future of Development is Hybrid

“Total Effort: 45 Minutes. Traditional Estimate: 2-4 Days. The multiplier is real.”



We are not replacing developers; we are equipping them with a scalable, intelligent workforce. The combination of Swarm Orchestration for breadth and Direct Assistance for depth is the new standard for velocity.